Mini Project

```
Start coding or generate with AI.
```

```
import pandas as pd

data = {
    'Name': ['A', 'B', 'C', 'D', 'E'],
    'Math':[75, 69, 82, 55, 78],
    'Science':[89, 71, 96, 69, 85],
    'English': [85, 79, 92, 65, 88]
}

df = pd.DataFrame(data)

print(df)
```

```
  Name  Math  Science  English
0    A    75       89       85
1    B    69       71       79
2    C    82       96       92
3    D    55       69       65
4    E    78       85       88
```

```
import numpy as np
import pandas as pd

marks = df[['Math', 'Science', 'English']].values

average_marks = np.mean(marks, axis=1)

df['Average'] = average_marks

print(df)
```

```
  Name  Math  Science  English    Average
0    A    75       89       85  83.000000
1    B    69       71       79  73.000000
2    C    82       96       92  90.000000
3    D    55       69       65  63.000000
4    E    78       85       88  83.666667
```

```
def grade(avg):
    if avg >= 90:
        return 'A'
    elif avg >= 75:
        return 'B'
```

```
    elif avg >= 60:
        return 'C'
    elif avg >= 50:
        return 'D'
    else:
        return 'F'
df['Grade'] = df['Average'].apply(grade)
df
```

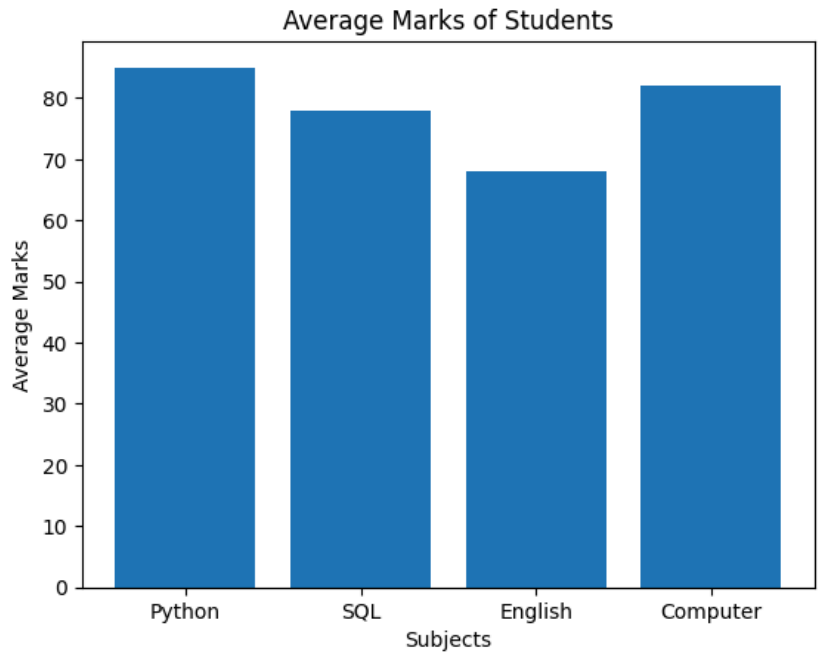
	Name	Math	Science	English	Average	Grade
0	A	75	89	85	83.000000	B
1	B	69	71	79	73.000000	C
2	C	82	96	92	90.000000	A
3	D	55	69	65	63.000000	C
4	E	78	85	88	83.666667	B

Next steps: Generate code with `df` New interactive sheet

```
import matplotlib.pyplot as plt

subjects = ["Python", "SQL", "English", "Computer"]
average_marks = [85, 78, 68, 82]

plt.figure()
plt.bar(subjects, average_marks)
plt.xlabel("Subjects")
plt.ylabel("Average Marks")
plt.title("Average Marks of Students")
plt.show()
```




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